THE HONG KONG UNIVERSITY OF SCIENCE AND TECHNOLOGY (GUANGZHOU)

THE HONG KONG UNIVERSITY OF SCIENCE AND TECHNOLOGY

AIAA 2290: Ethics, Privacy and Security in AI

Privacy Across AI Domains: Natural Language Processing

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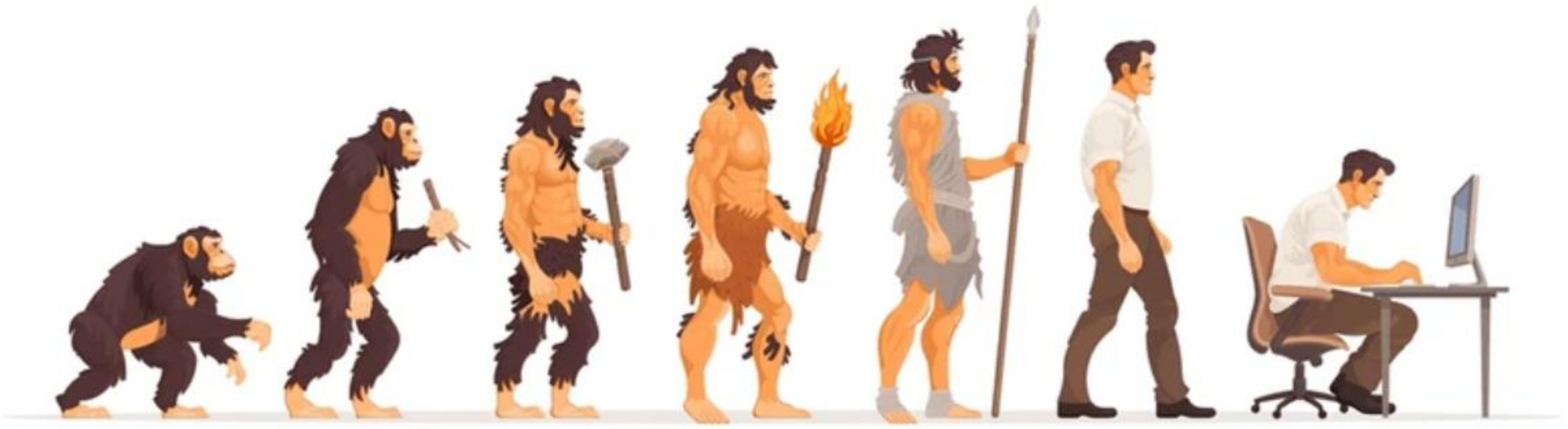
The Hong Kong University of Science and Technology (Guangzhou)

2025 Fall

Outline

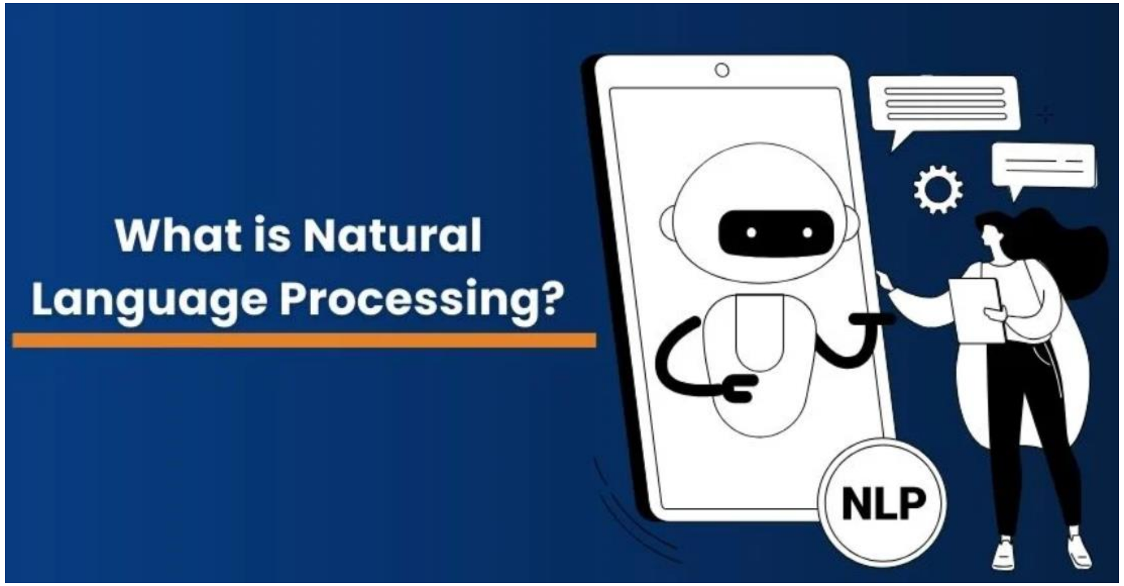
What is Natural Language?

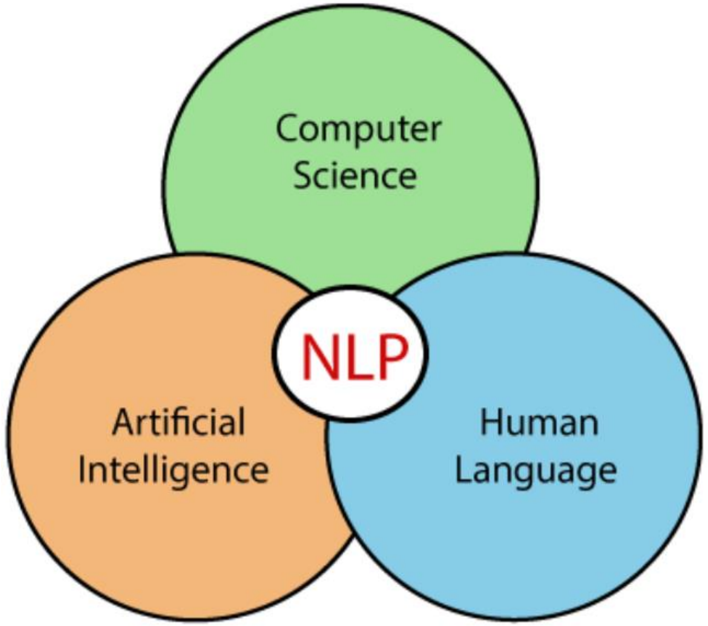


- Human intelligence and civilization
- The carrier of thought
- Communication and expressing emotions

What is NLP?

Natural Language Processing



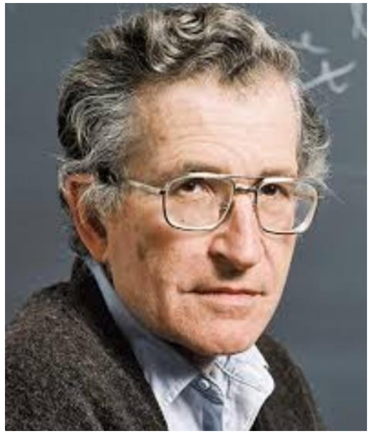


NLP aims to equip machines with the ability to **comprehend, interpret, and generate human language**, enabling applications like machine translation, speech recognition, and conversational AI.

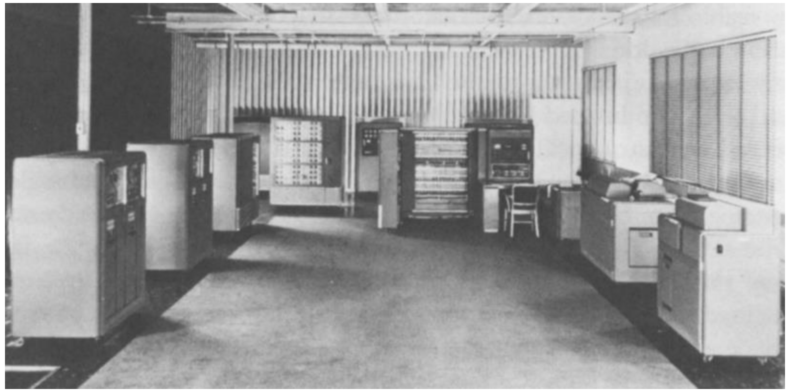
Milestones in the Development of NLP

1. Early Stage (1950s - 1960s)

Alan Turing



Noam Chomsky



IBM 701 Machine Translation

- **1950: Turing Test**

  Can a machine exhibit intelligent behavior through language?

- **1954: IBM 701 Machine Translation**

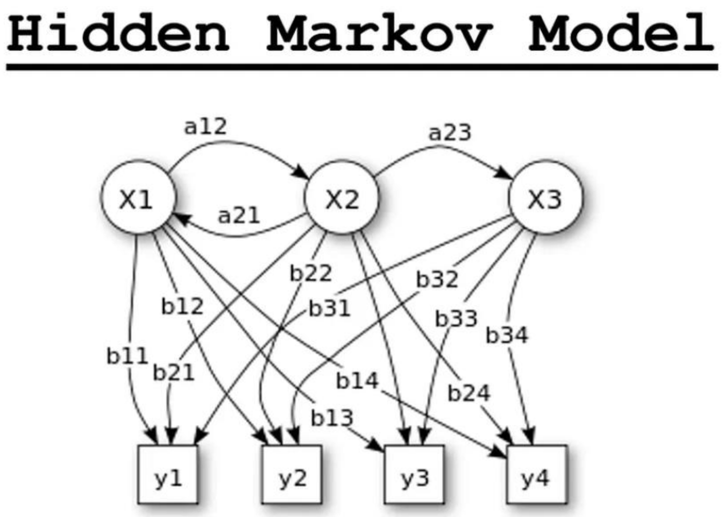
  First automatic machine translation system

- **1960: Chomsky's Linguistic Theory**

  Generative grammar theory

Milestones in the Development of NLP

2. The Rise of Statistical Methods (1970s - 1980s)



- **1970s: Introduction of Statistical Language Models**

Researchers began using **probability-based models** to analyze relationships between words. This marked a transition from rule-based systems to **data-driven approaches**.

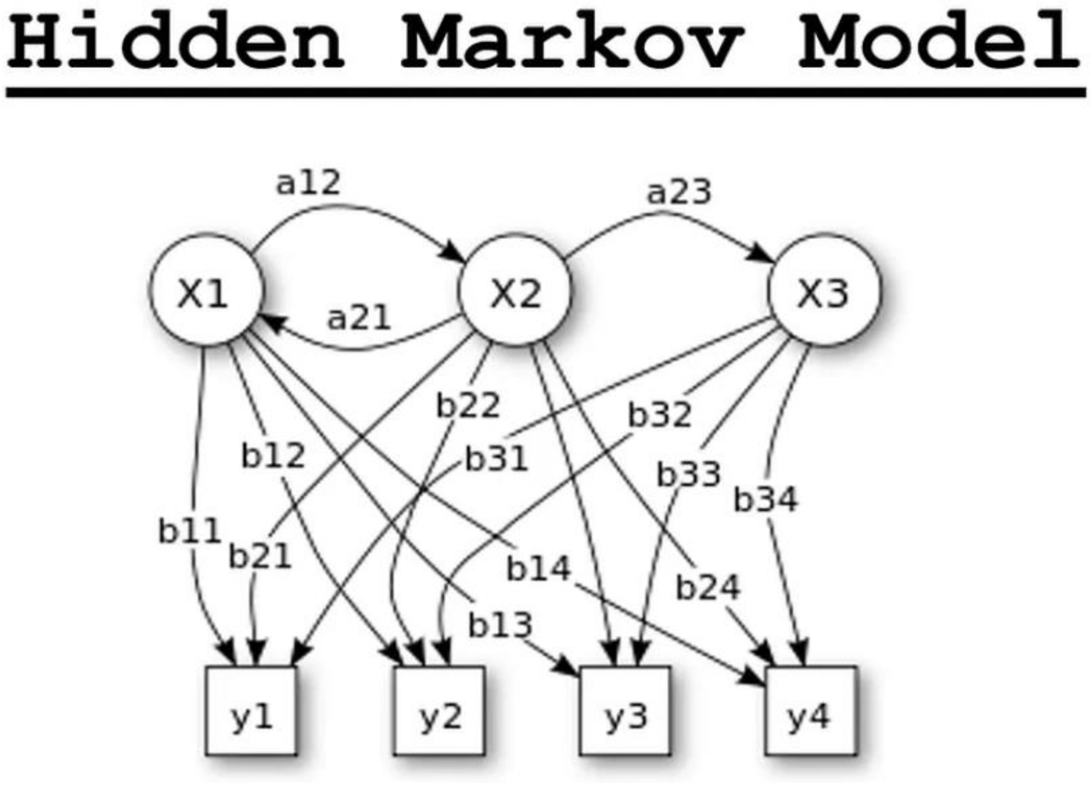
- **1980: Hidden Markov Models (HMM)**

part-of-speech tagging, speech recognition, and sequence prediction.

- **1988: Corpus-based Linguistics**

Researchers started using **large text corpora** to study language structures

2. The Rise of Statistical Methods (1970s - 1980s)



- **Hidden States (X1, X2, X3)**

  Represented by circles, these are **unobserved states**, such as grammatical structures, topics, or phonemes in NLP.

- **Observations (y1, y2, y3, y4)**

  Represented by squares, these are the **actual data**, like acoustic signals in speech recognition or words in POS tagging.

- **Transition Probabilities (a12, a21, a23, etc.)**

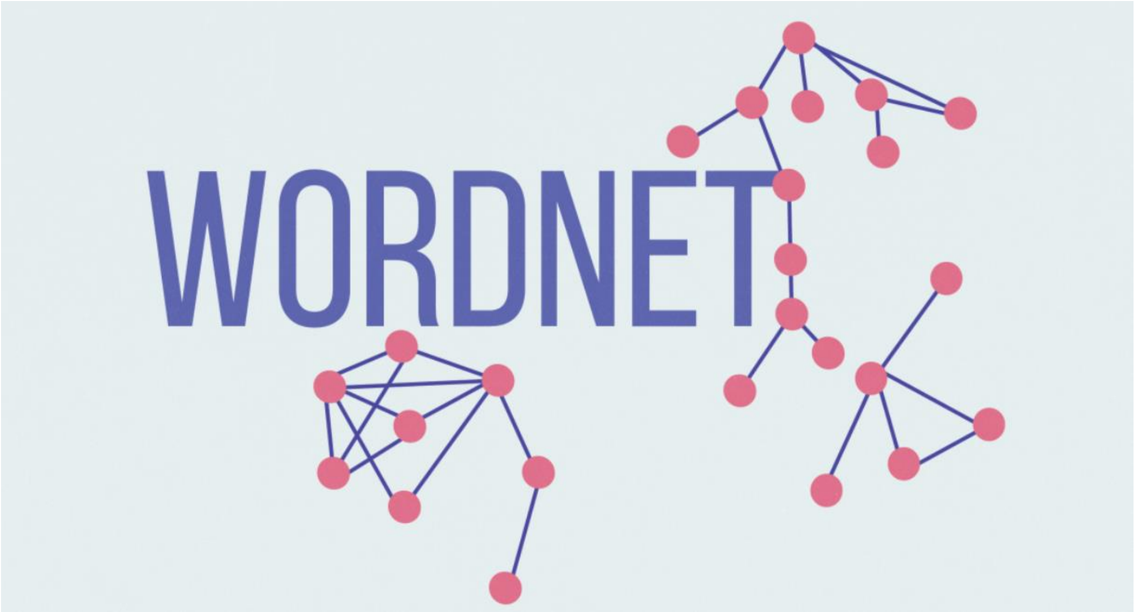
  Indicate the **chance of moving** from one hidden state to another, e.g., predicting phoneme sequences in speech.

- **Emission Probabilities (b11, b12, b21, etc.)**

  Define how likely an observation (y) comes from a hidden state (X), e.g., assigning a word to a specific POS tag.

Milestones in the Development of NLP

3. Application of Machine Learning (1990s - 2000s)



- **1992: Release of WordNet**

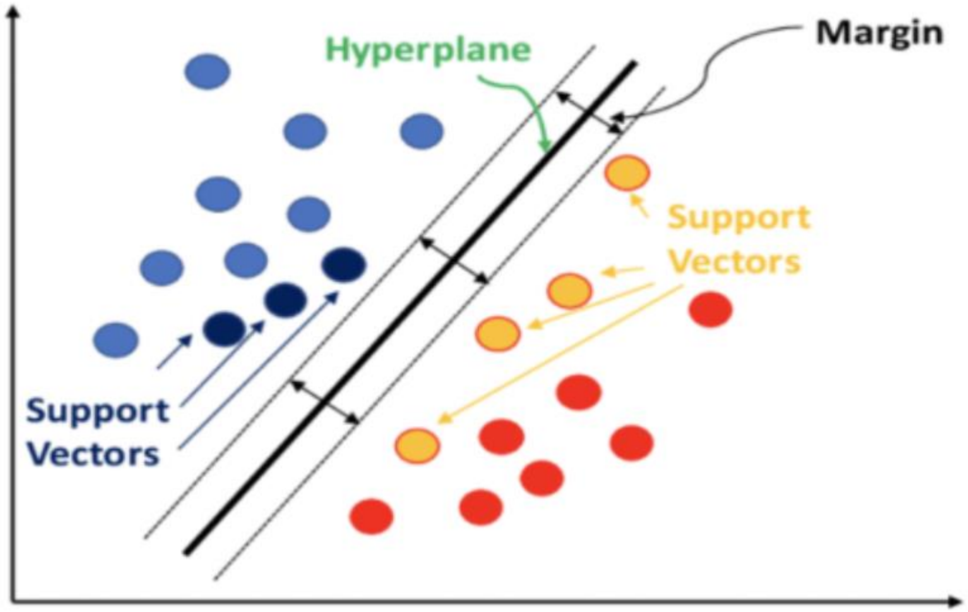
  A structured lexical database that became a foundational resource for NLP.

- **1993: Statistical Machine Translation (SMT)**

  Shifted from rule-based to **statistical** translation methods, enhancing translation accuracy.

Milestones in the Development of NLP

3. Application of Machine Learning (1990s - 2000s)



- **1998: Support Vector Machines (SVM) for Text Classification**

  A powerful algorithm that significantly improved **text classification** performance.

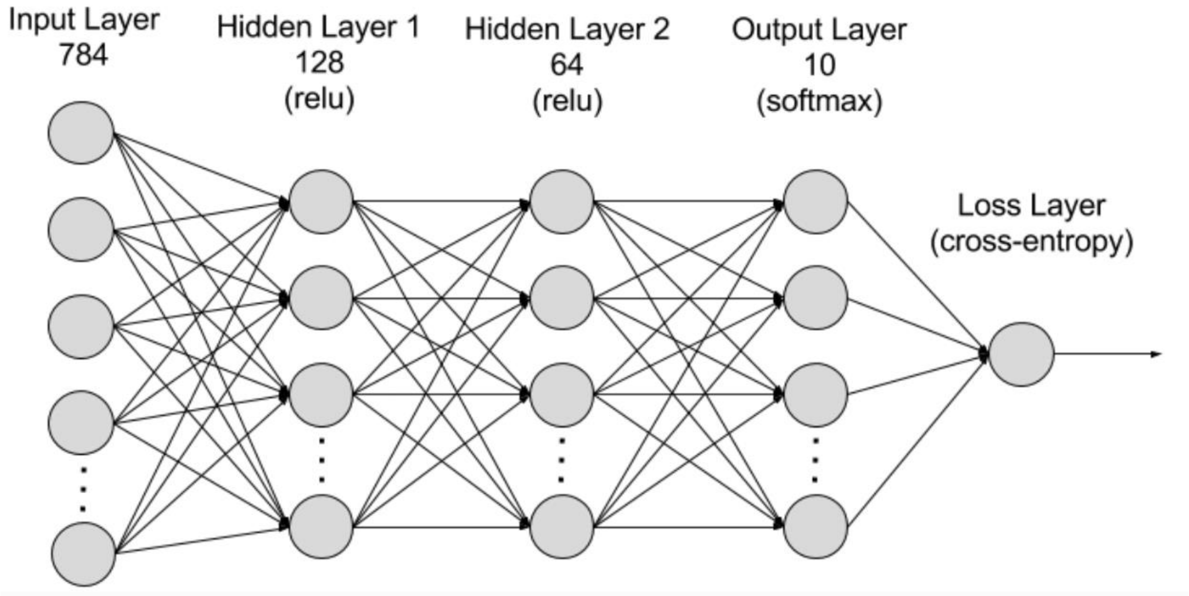
- **2003: Conditional Random Fields (CRF) for Sequence Labeling**

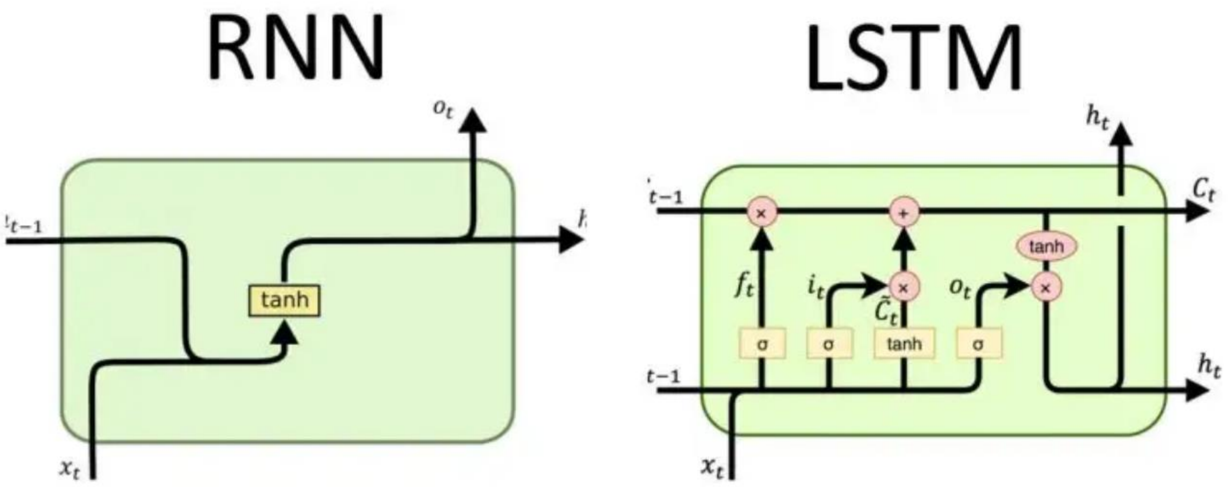
  Became essential for **sequence labeling** tasks like named entity recognition (NER) and part-of-speech tagging.

Milestones in the Development of NLP

4. The Era of Deep Learning (2010s - Present)

- **RNN**
- **LSTM**



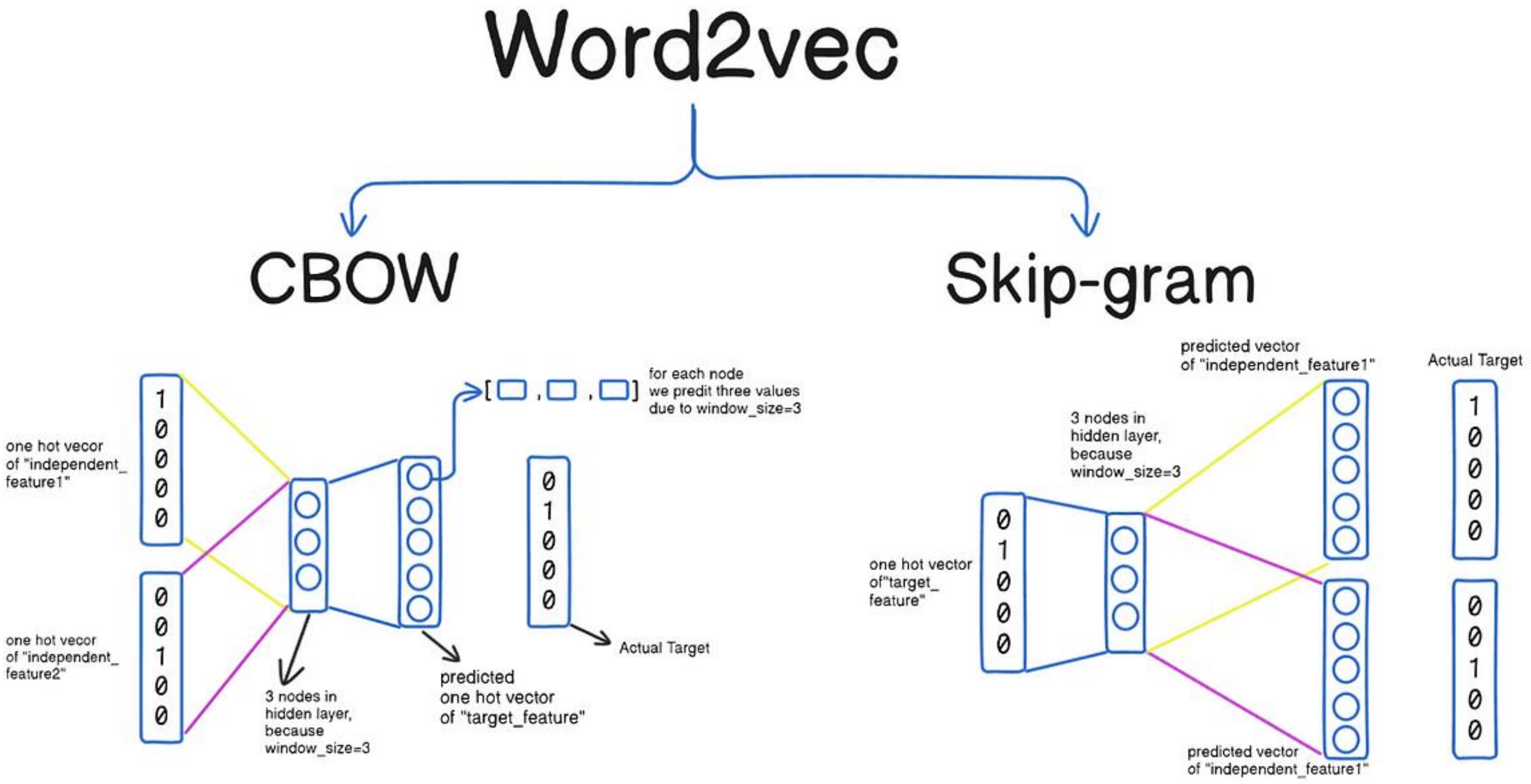


1. RNNs are designed to process **sequential data**.
2. They consist of **input, hidden, and output layers**, where the hidden layer captures dependencies between words over time.

1. LSTMs, an advanced version of RNNs, were developed to overcome the **vanishing gradient problem**, allowing models to remember long-term dependencies.
2. Google introduced **Sequence-to-Sequence (Seq2Seq) models** using LSTMs.

Milestones in the Development of NLP

4. The Era of Deep Learning (2010s - Present)



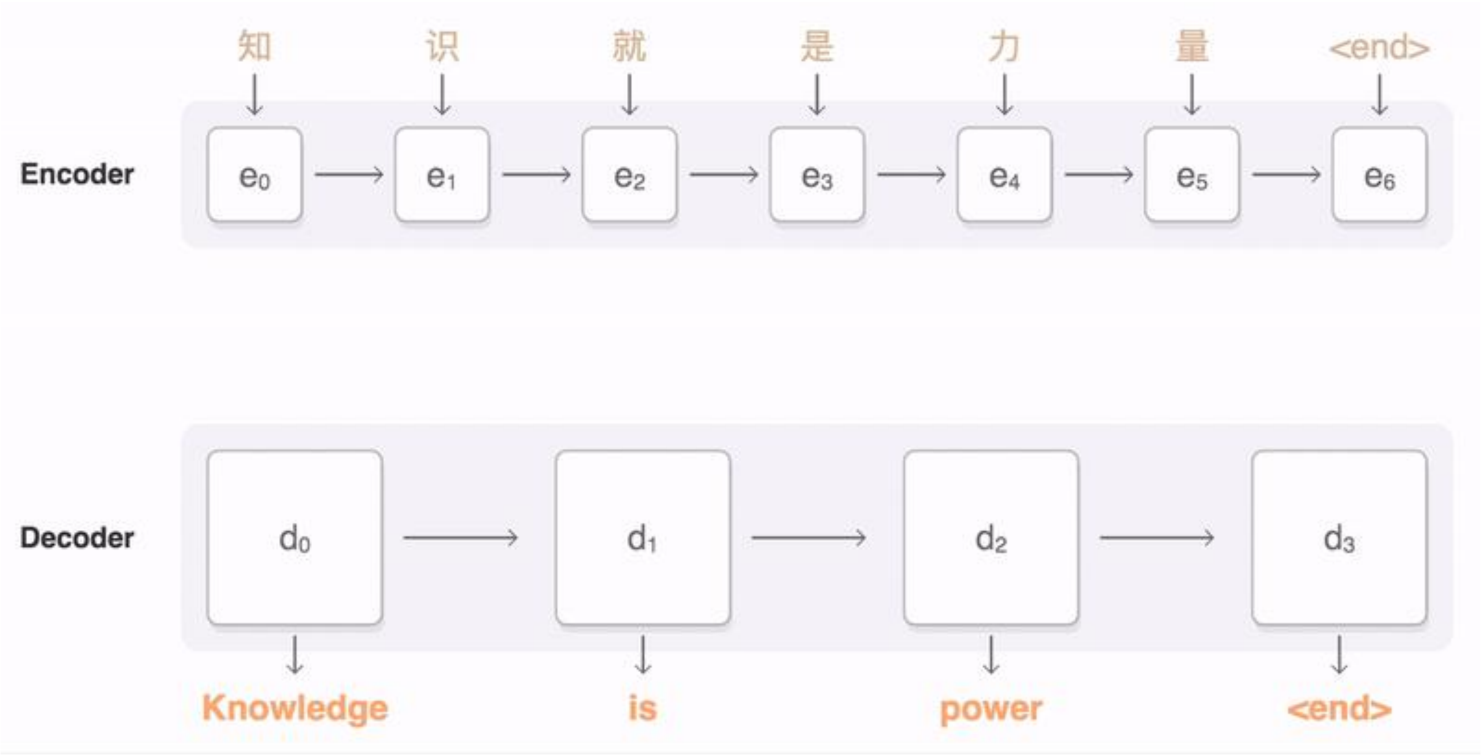
1. The **CBOW** model predicts the target word based on the context words surrounding it.

2. The **Skip-gram** model works in the opposite way: it predicts the surrounding context words based on a target word

Milestones in the Development of NLP

4. The Era of Deep Learning (2010s - Present)

- **Seq2Seq:** one sequence of text to be transformed into another

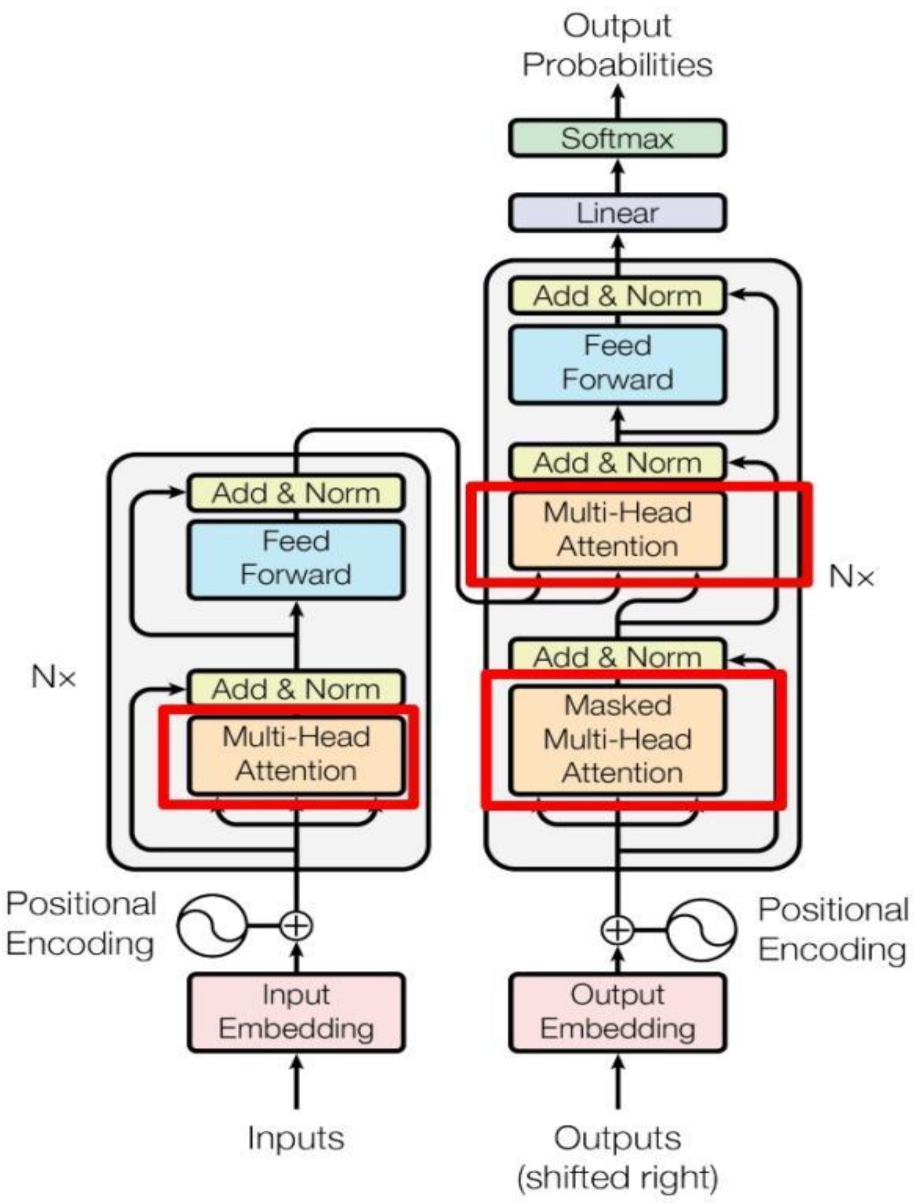


Milestones in the Development of NLP

4. The Era of Deep Learning (2010s - Present)

Transformer

The Transformer consists of an **encoder** and a **decoder**, utilizing **multi-head attention** to capture different levels of features and employing residual connections and a feed-forward network (FFN) for information propagation and transformation.



Milestones in the Development of NLP

4. The Era of Deep Learning (2010s - Present)

- **2018: BERT (Bidirectional Encoder Representations from Transformers) Released**

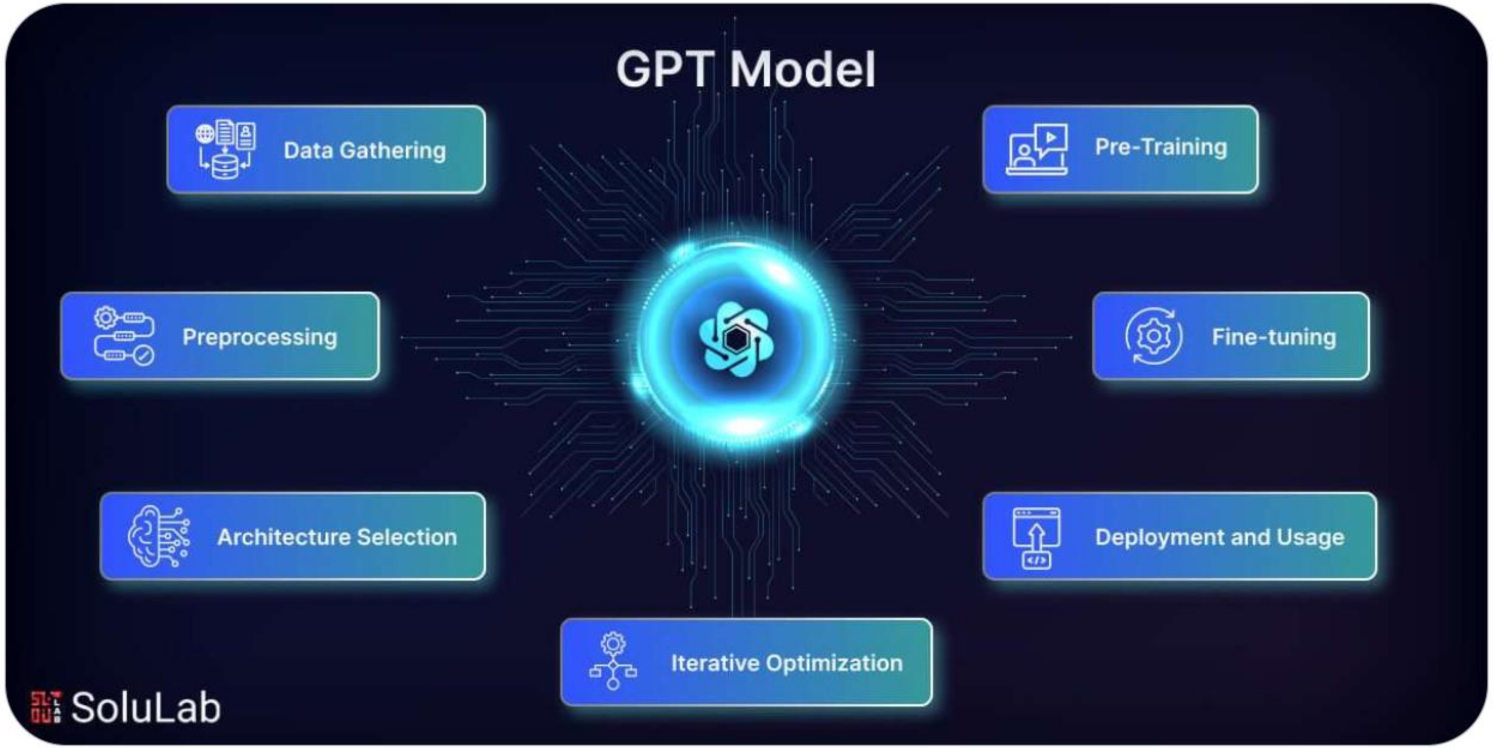
- BERT is a **bidirectional Transformer model** designed to **capture contextual information from both the left and right of a given word** in a sentence.

- Unlike previous models that processed text **left-to-right (like GPT) or right-to-left**, BERT **learns bidirectional dependencies** using the **masked language modeling (MLM) objective**.

Milestones in the Development of NLP

4. The Era of Deep Learning (2010s - Present)

- **2018: GPT (Generative Pre-trained Transformer) Released**



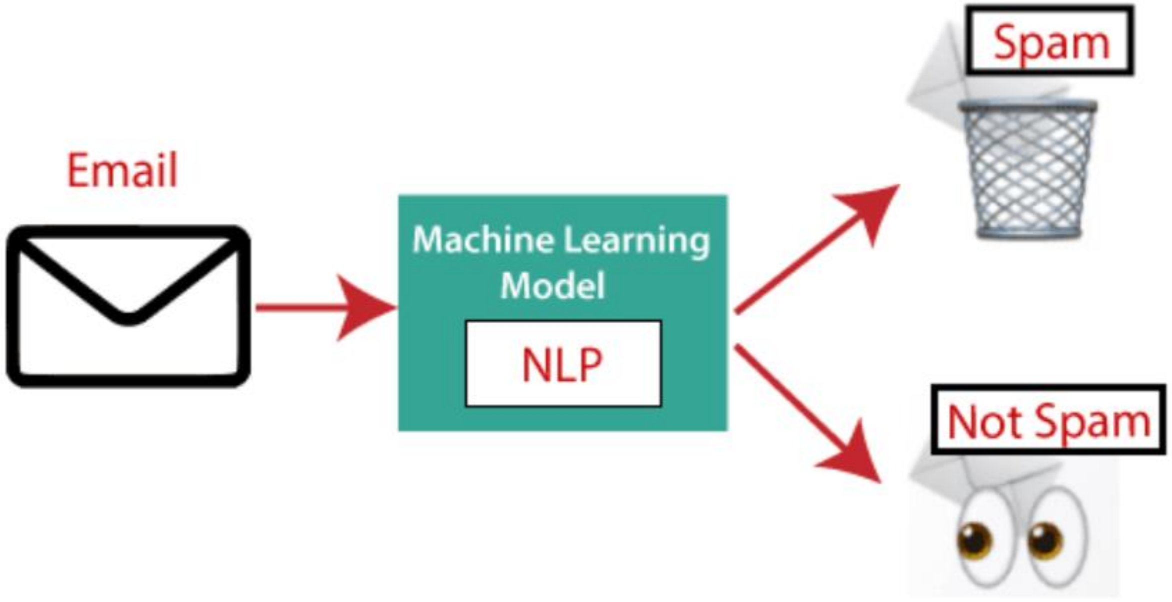
- Developed by **OpenAI**, GPT introduced the **pre-trained language model paradigm**, where a model is **trained on massive text corpora and fine-tuned for specific tasks**.
- GPT follows a **left-to-right, autoregressive** approach, making it highly effective for **text generation tasks**.

Applications of NLP

Question Answering

- Question Answering systems are designed to **automatically understand and answer human queries** in natural language.

Spam detection



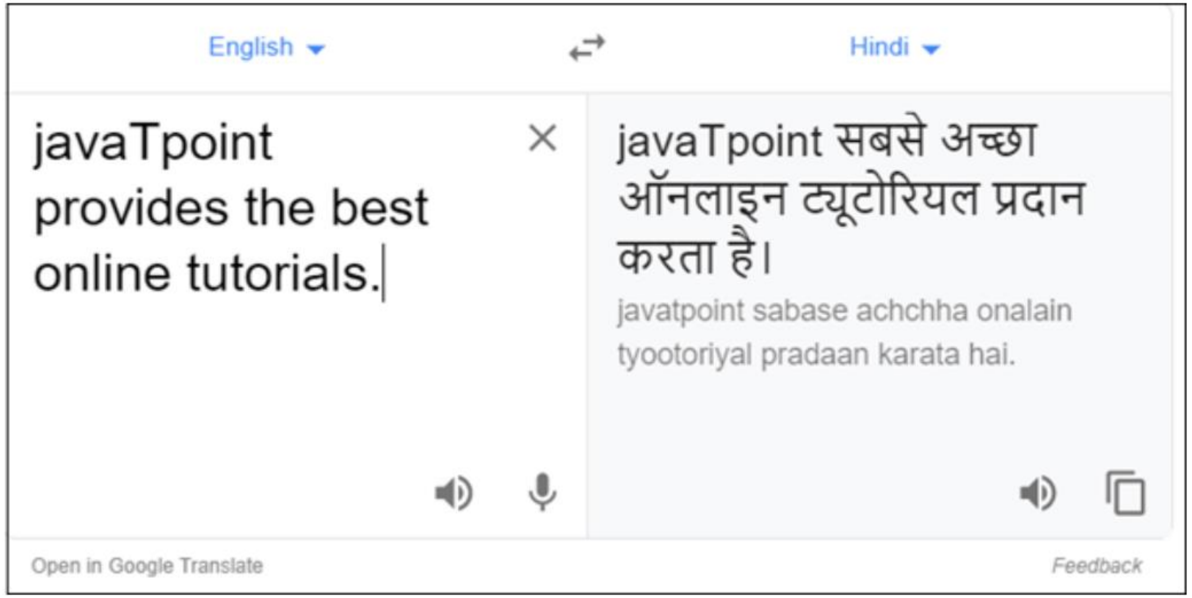
- Spam detection helps filter out **unwanted emails and messages** from a user's inbox.

Applications of NLP

Sentiment Analysis

Sentiment Analysis is also known as opinion mining. It is used on the web to analyse the attitude, behaviour, and emotional state of the sender.

Machine translation



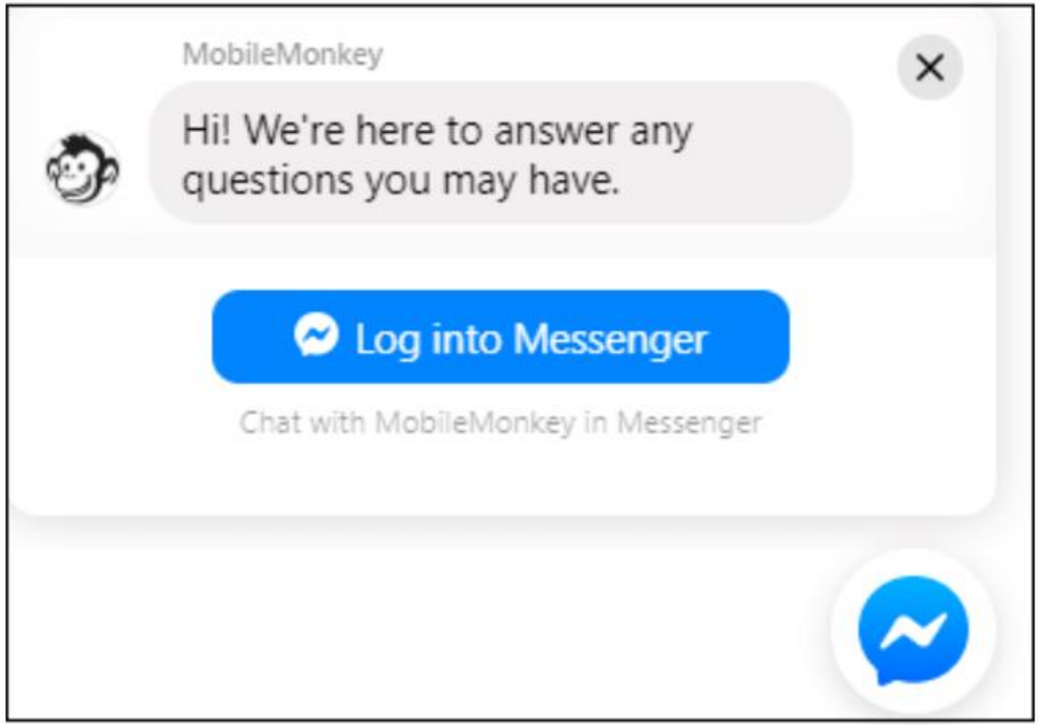
Machine translation is used to translate text or speech from one natural language to another natural language.

Applications of NLP

Spelling correction

Speech recognition

Chatbot

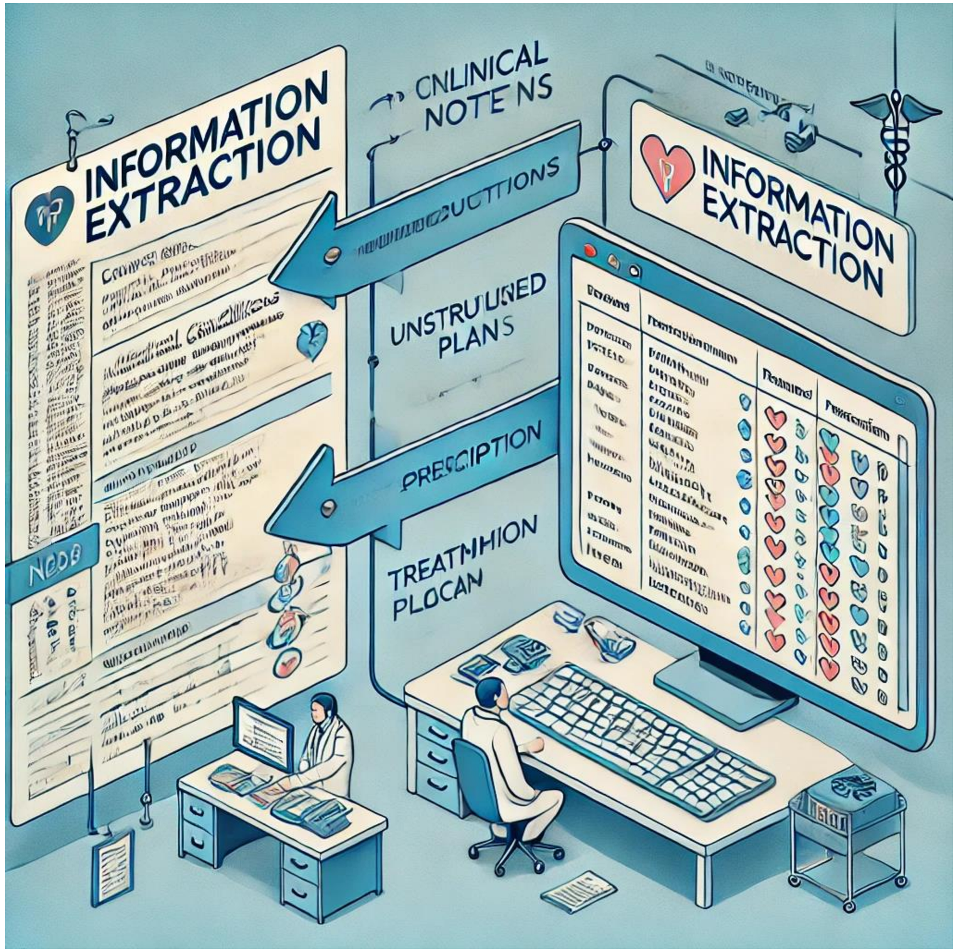


Chatbots are widely used in customer service to provide instant responses to common queries.

For instance, many banks use chatbots to help customers check their account balance, report lost cards, and inquire about transaction history.

Applications of NLP

Information extraction



IE is the process of automatically extracting structured information from unstructured text.

Privacy Concerns in NLP

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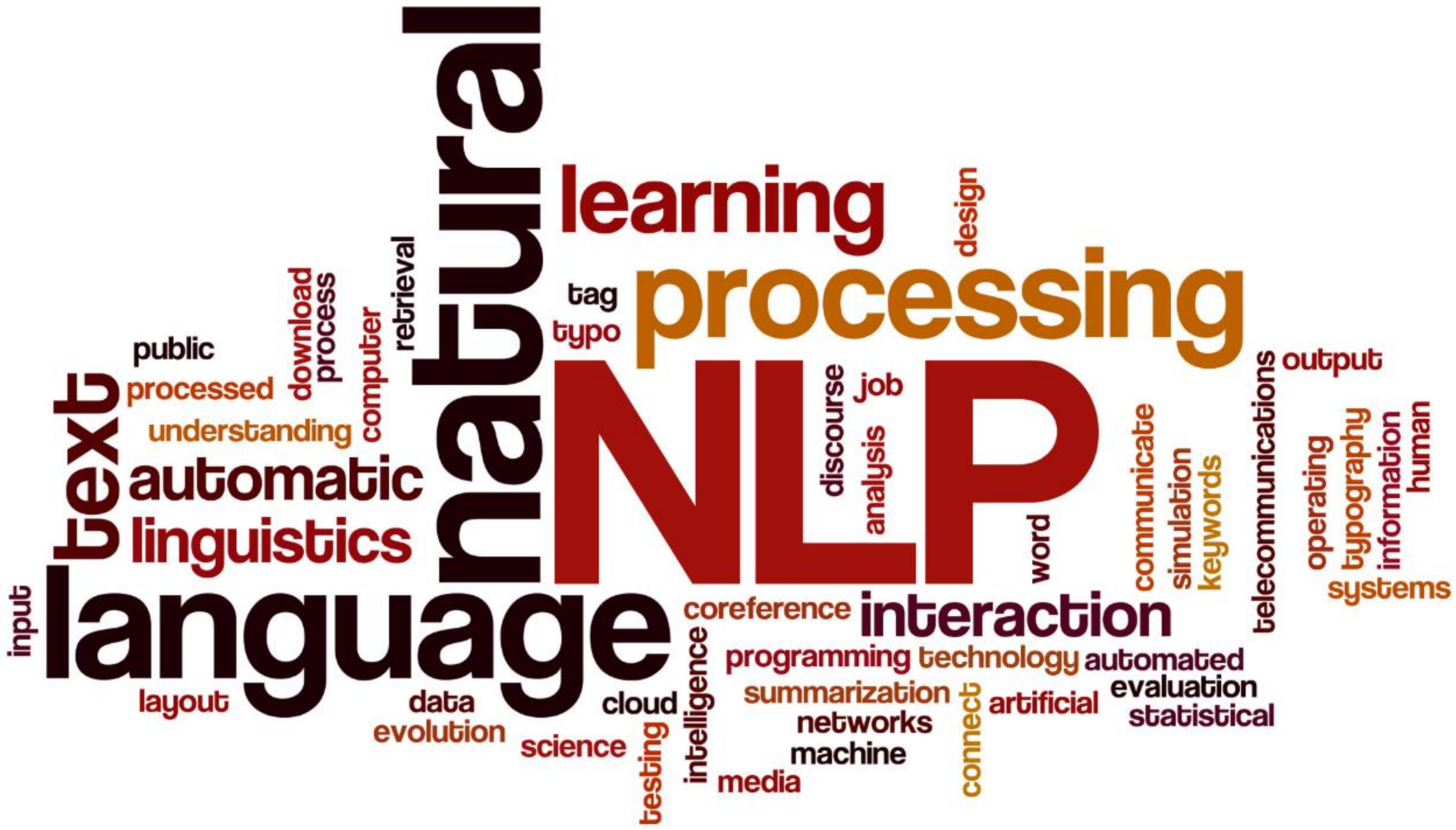
NLP Models Depend on Large Text Datasets

Sensitivity of Text Data



- NLP models are trained on vast amounts of text, which can include **social media posts, emails, chat logs, and even medical records**.

- Many of these texts contain **personally identifiable information (PII)** and other **highly sensitive details**. This includes **names, locations, health data, financial transactions, and behavioral patterns**.

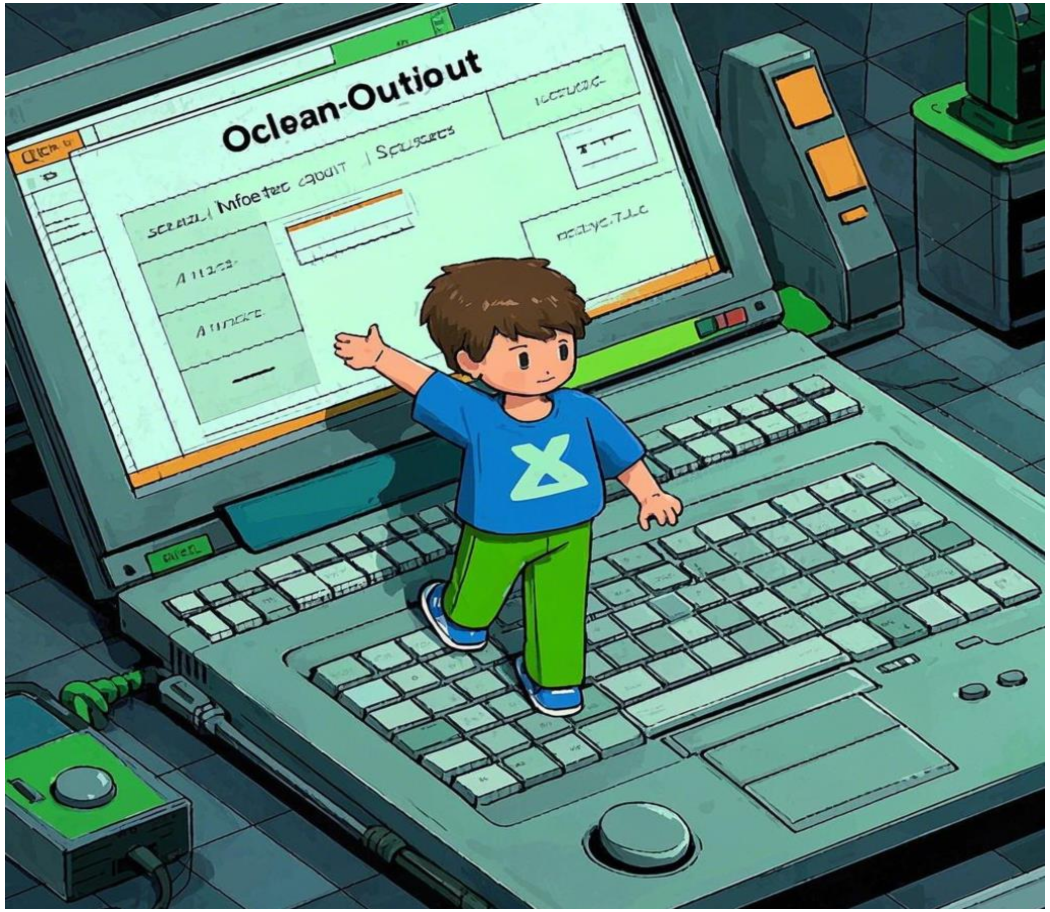
Potential Risks of Privacy Breaches

Models May Memorize Private Information from Training Data

During inference, these models can unintentionally **generate text that resembles private data**, exposing **personal details, confidential emails, or financial records**.

NLP Models Depend on Large Text Datasets

Model Output as a Source of Inference Attacks



Inference attacks exploit a model's generated responses to extract sensitive training data.

Attackers can **query the model repeatedly**, attempting to reconstruct hidden training data.

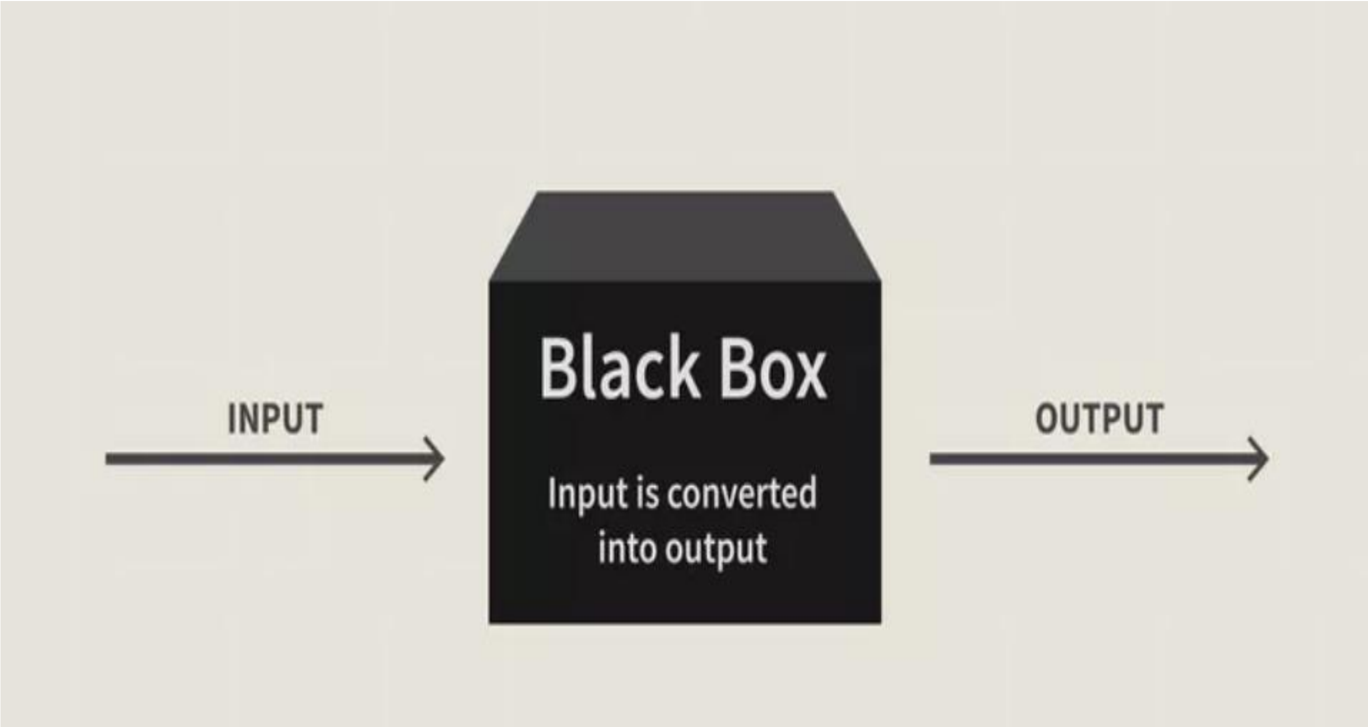
Privacy Concerns in NLP

NLP Models Depend on Large Text Datasets

Video Website: https://www.youtube.com/watch?v=wY109DfbbCE

NLP Models Depend on Large Text Datasets

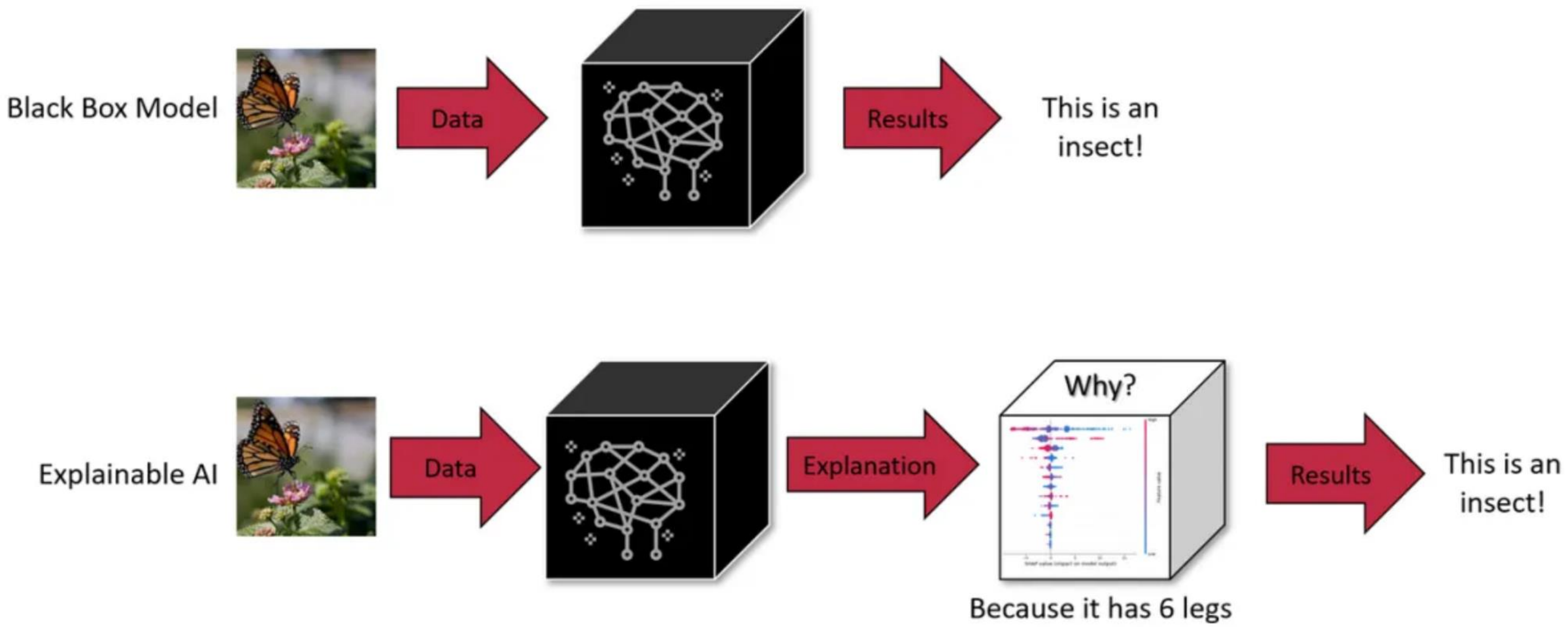
Black-box Models and Risks



- **The internal workings are not fully transparent.**
- **Not fully understand how the model processes inputs** to generate outputs.
- Challenges in **debugging, ensuring fairness, and preventing data leakage**.

NLP Models Depend on Large Text Datasets

Black-box Models and Risks

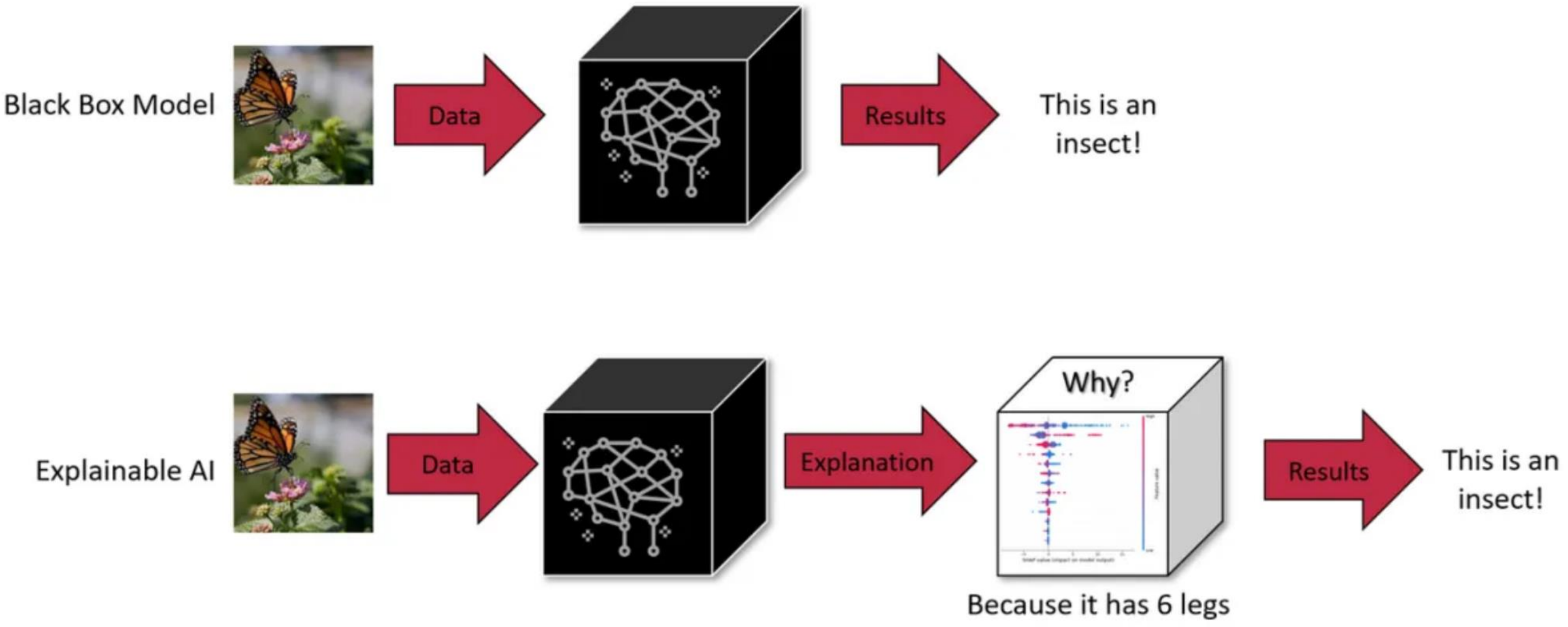


Privacy Risks of Black-box NLP Models

- Unintended Information Disclosure
- Bias & Ethical Issues
- Security Vulnerabilities

NLP Models Depend on Large Text Datasets

Black-box Models and Risks



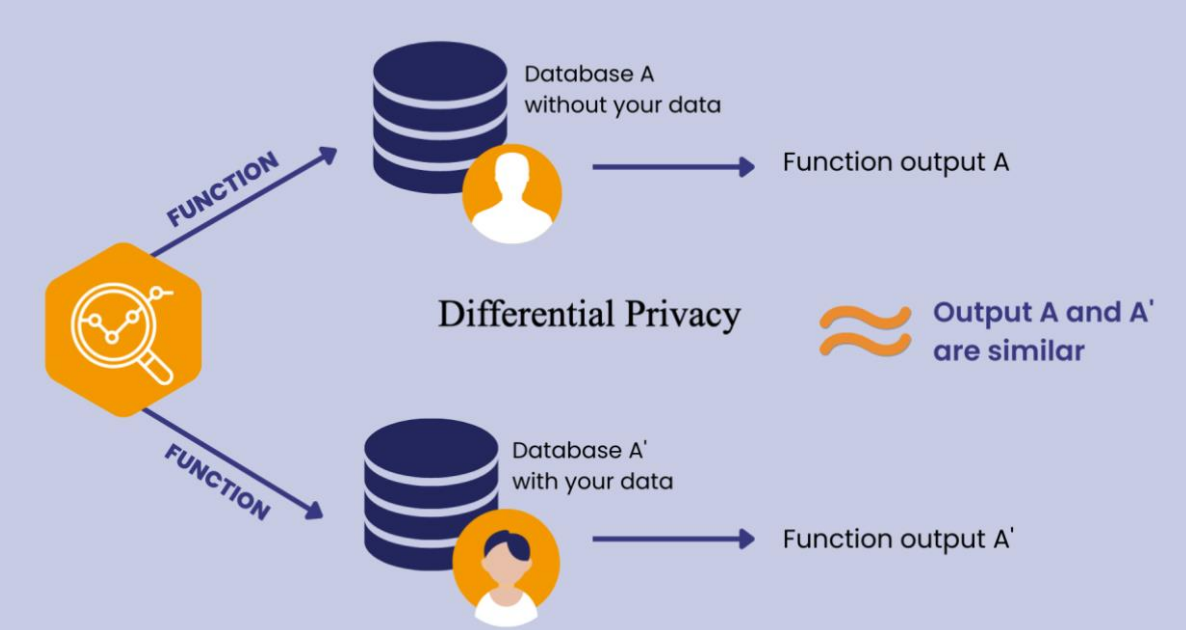
Why Model Interpretability Matters?

- Better Debugging & Trust
- Privacy & Compliance
- Fair & Ethical AI

Please write down your understanding of black box models and the risks of black box models, and try to write some methods that you think are effective in improving the interpretability of the model.

Privacy Protection Techniques in NLP

Differential Privacy (DP)



- It ensures that **adding or removing a single individual's data** from a dataset **does not significantly change** the output of a query.
- This prevents attackers from determining whether a specific person's data is included in the dataset.

- **Adding Noise**: DP achieves privacy by introducing **random noise** (typically **Laplace or Gaussian noise**) to query results.
- **Statistical Similarity**: The noisy outputs make it **difficult for attackers to infer sensitive information**, ensuring that individual data points remain hidden.

Applications of DP in LLMs

A. DP in Training

- Differentially Private Stochastic Gradient Descent (DP-SGD)

Social Media Analysis:
On social media platforms, user private messages and interactions may be collected. By using DP-SGD (random noise is added to the gradients before they are applied to the model), user privacy can be protected while training sentiment analysis or recommendation algorithms.

Applications of DP in LLMs

B. DP in Inference

- Privacy-Aware Text Generation
- Output Clipping

User

I have been experiencing frequent headaches recently, accompanied by blurred vision. My blood pressure is measured at 160/100, and I live in a certain community in **Pudong, Shanghai.**

Privacy Protection Techniques in NLP

Applications of DP in LLMs

Without DP

You may have hypertension (Case ID: SH-2023-12345). It is recommended that you immediately contact the **XX Street Community** Health Center in **Pudong**, Shanghai, and book an appointment with Dr. Zhang XX.

With DP

Your current blood pressure level requires attention. It is recommended that you contact a local healthcare facility in your community for further examination as soon as possible.

Privacy Protection Techniques in NLP

Applications of DP in LLMs

B. DP in Inference

- Privacy-Aware Text Generation
- Output Clipping

User

My case number is BJ-2023-5678, involving an equity dispute with an internet company. According to Article 32 of the Company Law, do I have the right to review the company's articles of association?

Privacy Protection Techniques in NLP

Applications of DP in LLMs

Without DP

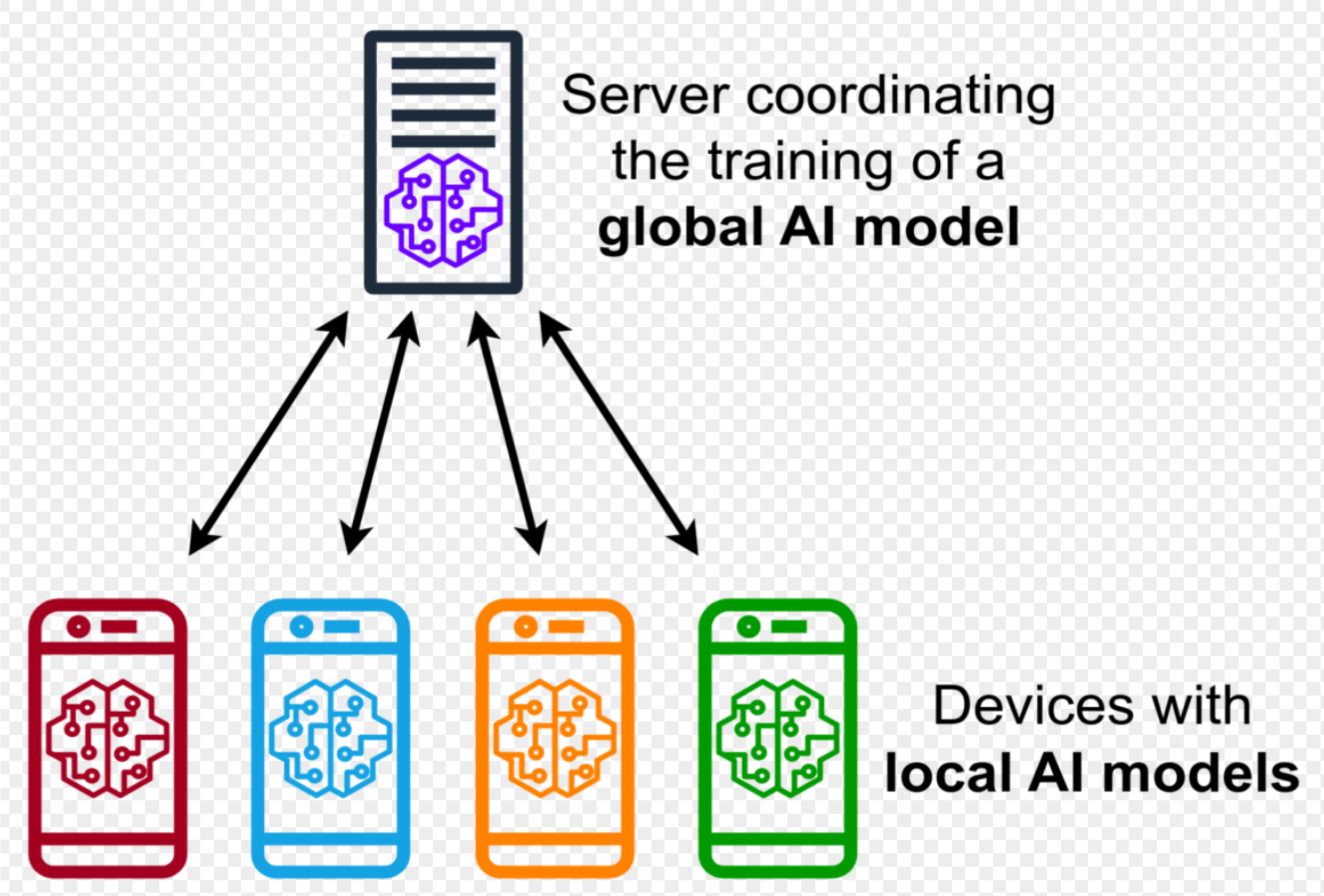
Based on your case BJ-2023-5678 and referring to Article 32 of the Company Law and a 2022 case from a Beijing court (Case No.: Jing 03 Min Zhong 1234), you have the right to request access to the company's articles of association.

With DP

According to relevant provisions of the Company Law, shareholders generally have the right to review the company's articles of association. It is recommended that you consult a legal professional for specific procedures.

Privacy Protection Techniques in NLP

Federated Learning



Federated Learning is a **distributed machine learning** approach that allows multiple devices to train a shared model **without transferring raw data** to a central server.

Privacy Protection Techniques in NLP

Federated Learning

- Cross-Device/Cross-Institution Collaborative LLM Training

  Personalized language models on users' mobile input methods (e.g., predictive text, autocorrection) are updated through federated learning without uploading sensitive content (e.g., passwords, chat records).

Privacy Protection Techniques in NLP

Federated Learning

- Cross-Device/Cross-Institution Collaborative LLM Training

- Joint Modeling of Sensitive Texts in Vertical Domains

  Banks collaboratively train anti-fraud models by analyzing semantic features in loan applications (e.g., contradictory statements) without sharing customer data across institutions.

Privacy Protection Techniques in NLP

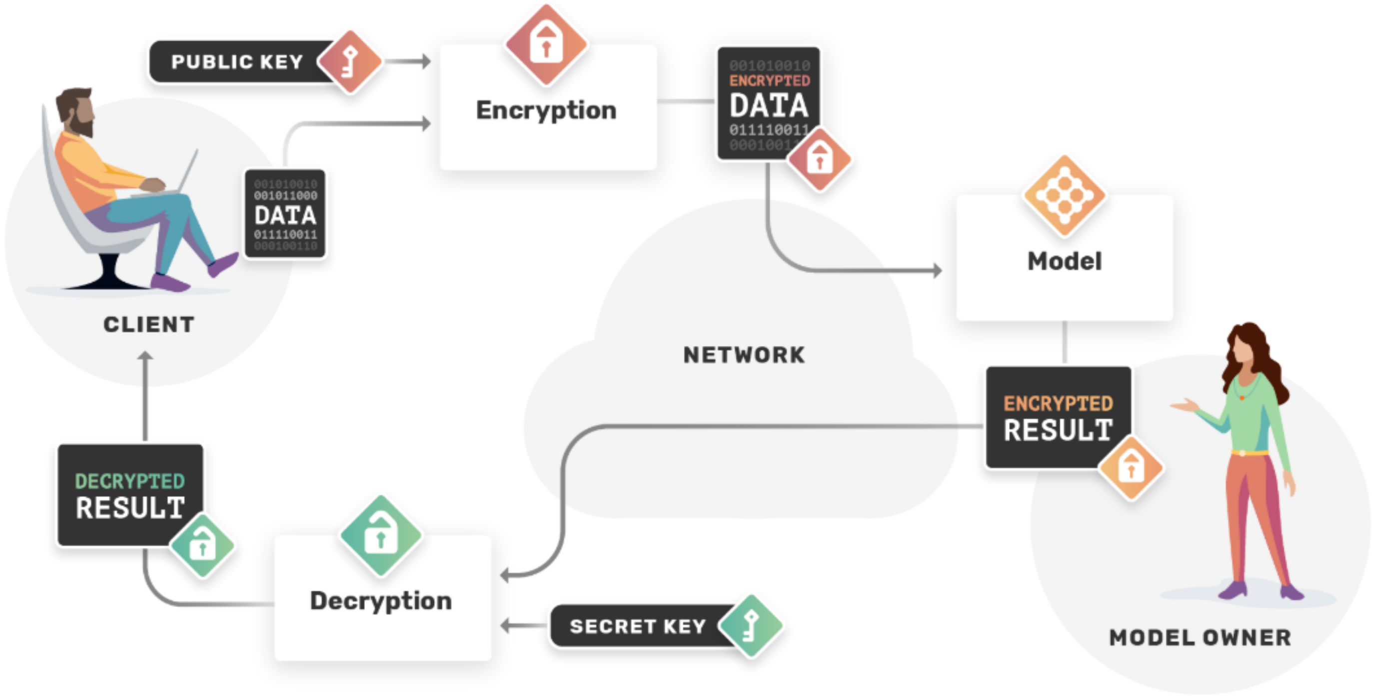
Federated Learning

- Cross-Device/Cross-Institution Collaborative LLM Training
- Joint Modeling of Sensitive Texts in Vertical Domains
- Real-Time LLM Fine-Tuning on Edge Devices

  Users fine-tune local LLMs through voice commands (e.g., preferred expressions), with federated aggregation improving the global model without uploading voice data.

Privacy Protection Techniques in NLP

Homomorphic Encryption



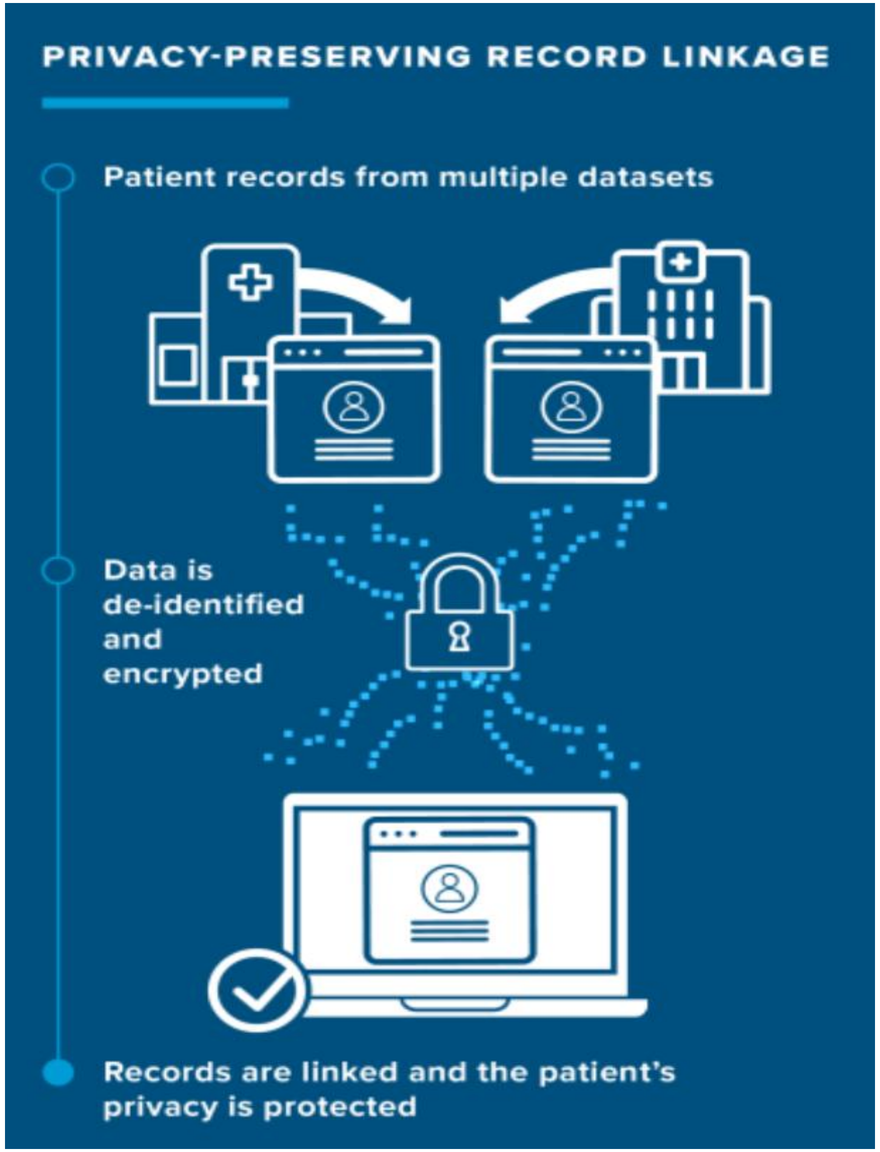
How Does Homomorphic Encryption Work?

1. Encryption
2. Computation on Encrypted Data
3. Encrypted Result
4. Decryption

Privacy Protection Techniques in NLP

Privacy-preserving Data De-identification

Data de-identification involves removing personal identifiers from datasets so that individuals cannot be identified directly or indirectly. After de-identification, the data can still be used for analysis without revealing sensitive personal information.



Privacy Protection Techniques in NLP

Privacy-preserving Data De-identification

lastname AT website DOT com

Private information has no one format

Die → Unalive

Language constantly changes

Privacy is context dependent

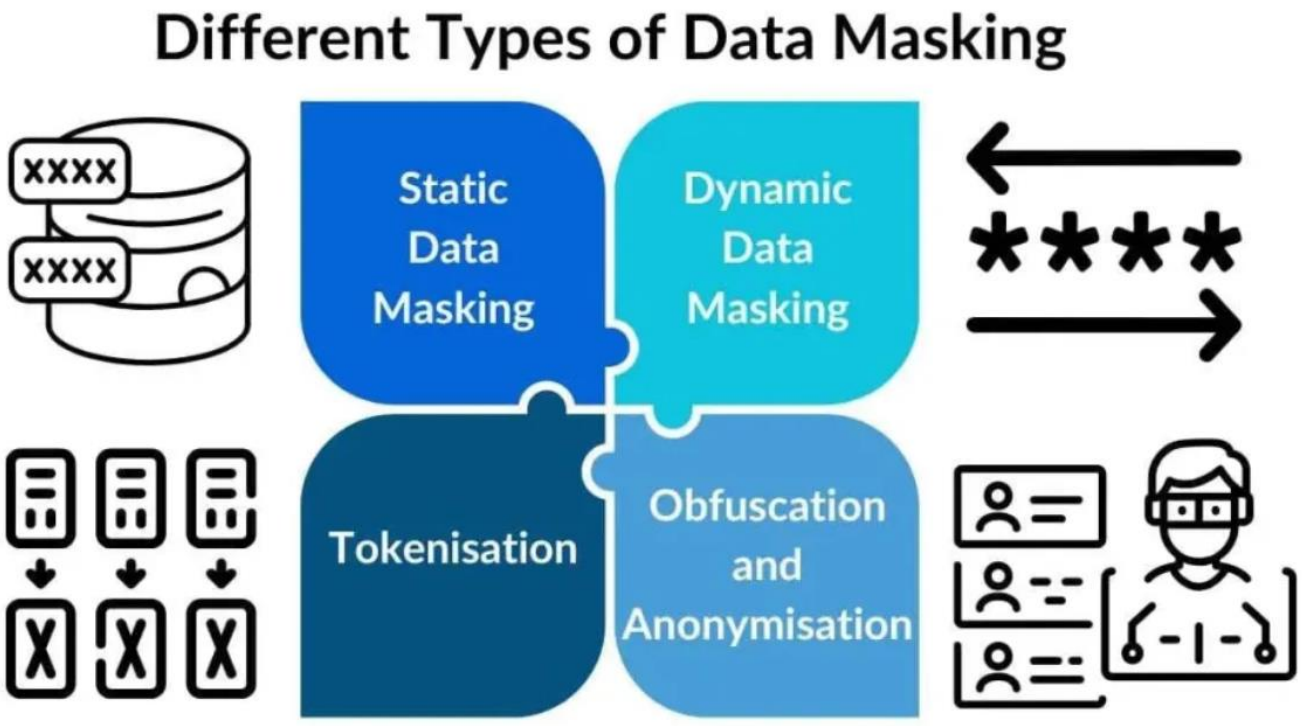
We're throwing **Bob** a surprise party

Privacy Protection Techniques in NLP

Privacy-preserving Data De-identification

A. Sensitive Information Masking in Preprocessing

B. Context-Aware De-identification in Text Generation



Privacy Protection Techniques in NLP

Privacy-preserving Data De-identification

C. Privacy Protection in Real-Time Conversational Systems

D. Protection Against Inference Attacks

Privacy Protection Techniques in NLP

Privacy-preserving Data De-identification

A. Sensitive Information Masking in Preprocessing

Raw Data

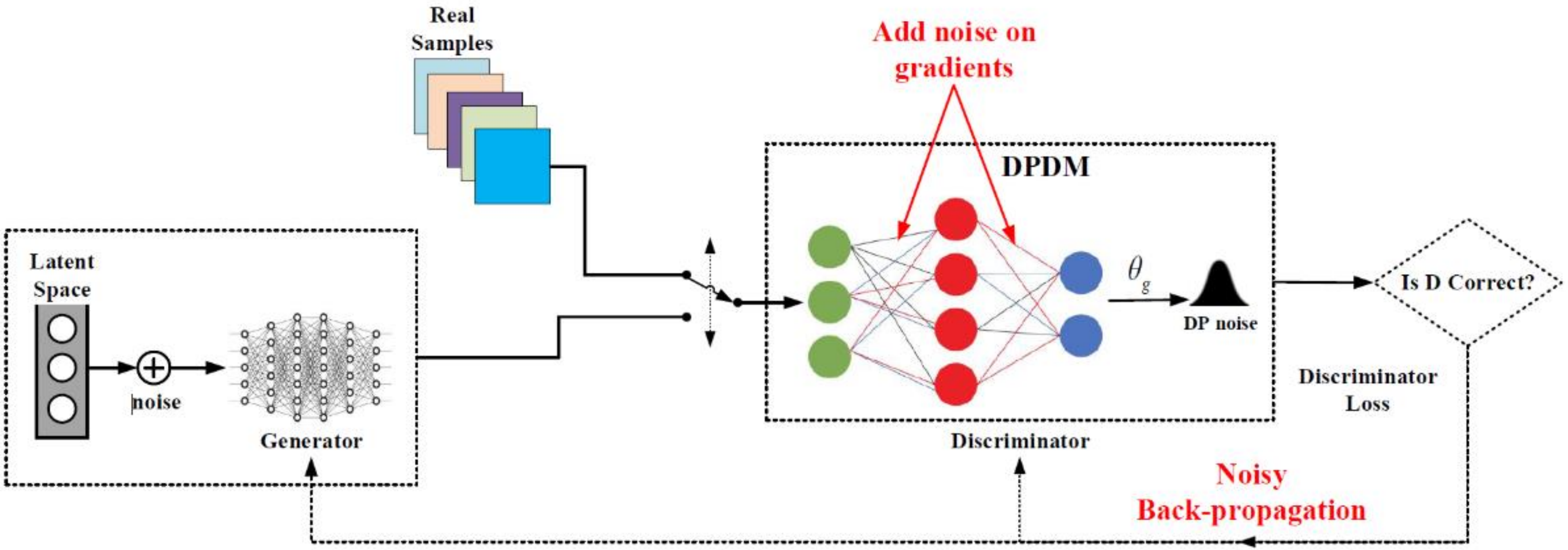
The patient, Zhang San, with ID number 310105199003041234 and phone number 13800138000, was admitted to Shanghai Sixth People's Hospital in May 2023. The chief complaint was persistent chest pain, and the electrocardiogram (ECG) showed ST-segment elevation. The diagnosis was acute myocardial infarction (ICD-10: I21.9).

Masked Data

Patient P_001, with ID number 310105********1234 and phone number 138****8000, was admitted to a top-tier hospital in Shanghai and diagnosed with a cardiovascular emergency (ICD code redacted).

Privacy Protection Techniques in NLP

Generative Adversarial Networks (GANs) for Privacy Protection



[1] Y. Liu, J. Peng, J. J. Q. Yu and Y. Wu, "PPGAN: Privacy-Preserving Generative Adversarial Network," *2019 IEEE 25th International Conference on Parallel and Distributed Systems (ICPADS)*, Tianjin, China, 2019, pp. 985-989, doi: 10.1109/ICPADS47876.2019.00150.

Privacy Protection Techniques in NLP

Generative Adversarial Networks (GANs) for Privacy Protection

A. Privacy-Preserving Synthetic Data Generation

B. Sensitive Feature Removal in Text Generation

C. Defense Against Inference Attacks

D. Data Augmentation and Few-Shot Learning

[1] Y. Liu, J. Peng, J. J. Q. Yu and Y. Wu, "PPGAN: Privacy-Preserving Generative Adversarial Network," *2019 IEEE 25th International Conference on Parallel and Distributed Systems (ICPADS)*, Tianjin, China, 2019, pp. 985-989, doi: 10.1109/ICPADS47876.2019.00150.

Privacy Protection Techniques in NLP

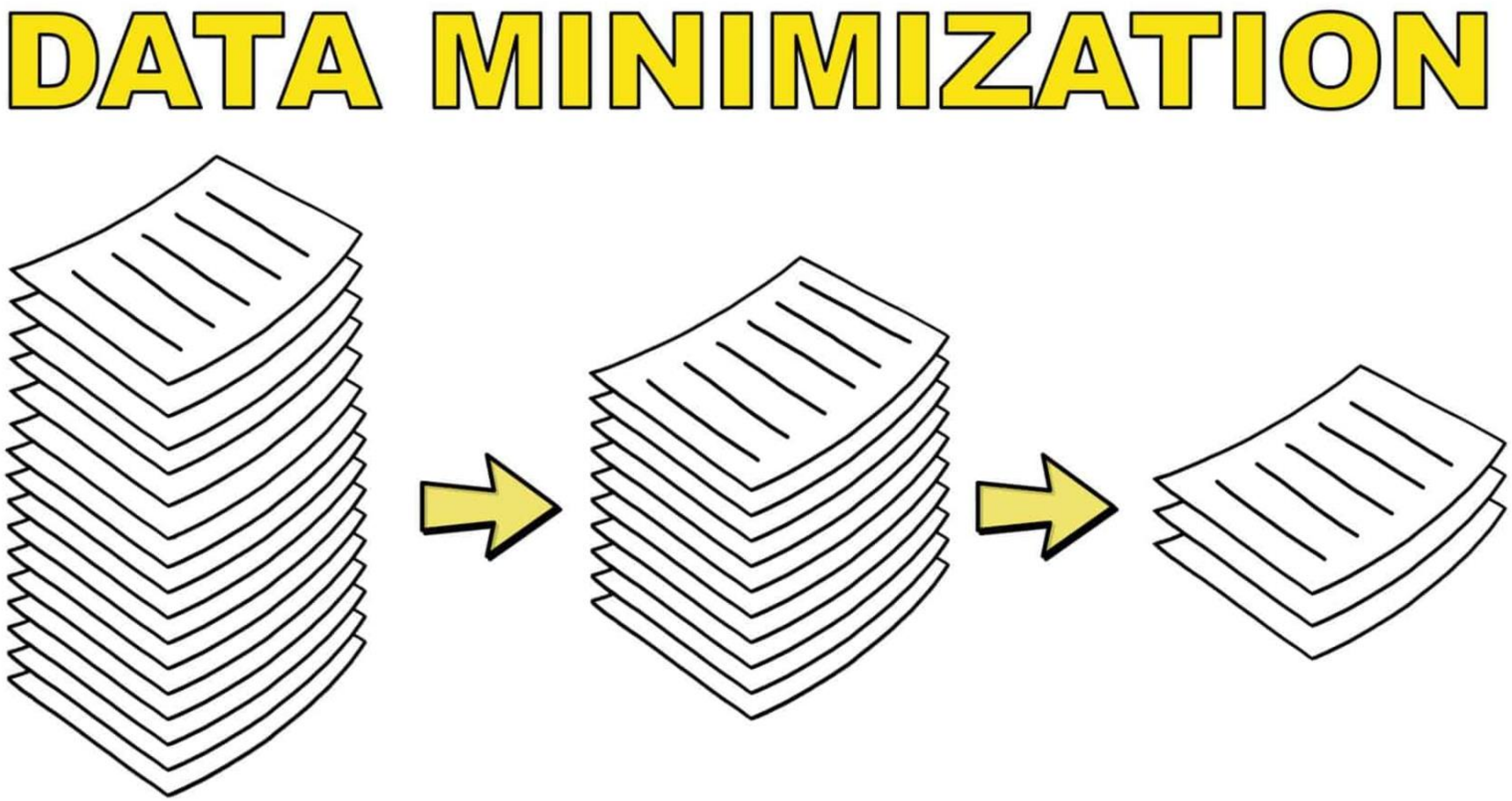
Generative Adversarial Networks (GANs) for Privacy Protection

Raw Data	Generated Data
Patient P_123, 50 years old, admitted to a top-tier hospital in March 2023, complaining of postprandial abdominal pain. Gastroscopy shows a gastric ulcer.	A middle-aged patient visited the hospital for recurring upper abdominal pain. Digestive endoscopy revealed chronic gastric mucosal lesions. Acid-suppressing treatment and follow-up were recommended.

[1] Y. Liu, J. Peng, J. J. Q. Yu and Y. Wu, "PPGAN: Privacy-Preserving Generative Adversarial Network," *2019 IEEE 25th International Conference on Parallel and Distributed Systems (ICPADS)*, Tianjin, China, 2019, pp. 985-989, doi: 10.1109/ICPADS47876.2019.00150.

Privacy Protection Techniques in NLP

Data Minimization



1.The large stack of papers on the left represents **raw, unfiltered data,** which may contain unnecessary or sensitive information.
2.The middle stack shows **filtered data with irrelevant information removed,** resulting in a reduced dataset.
3.The final, smaller stack on the right illustrates **the optimized dataset containing only the essential information**, helping to minimize storage and privacy risks.

Privacy Protection Techniques in NLP

Data Minimization

A. Limiting Data Collection Scope

B. Selective Feature Extraction

C. Data Truncation and Sampling

D. Minimizing Output Data Exposure

E. Data Retention Limitation

Privacy Protection Techniques in NLP

Data Minimization

An online bank offers intelligent customer service based on a Large Language model (LLM) to answer customers' questions about accounts, loans and investments.

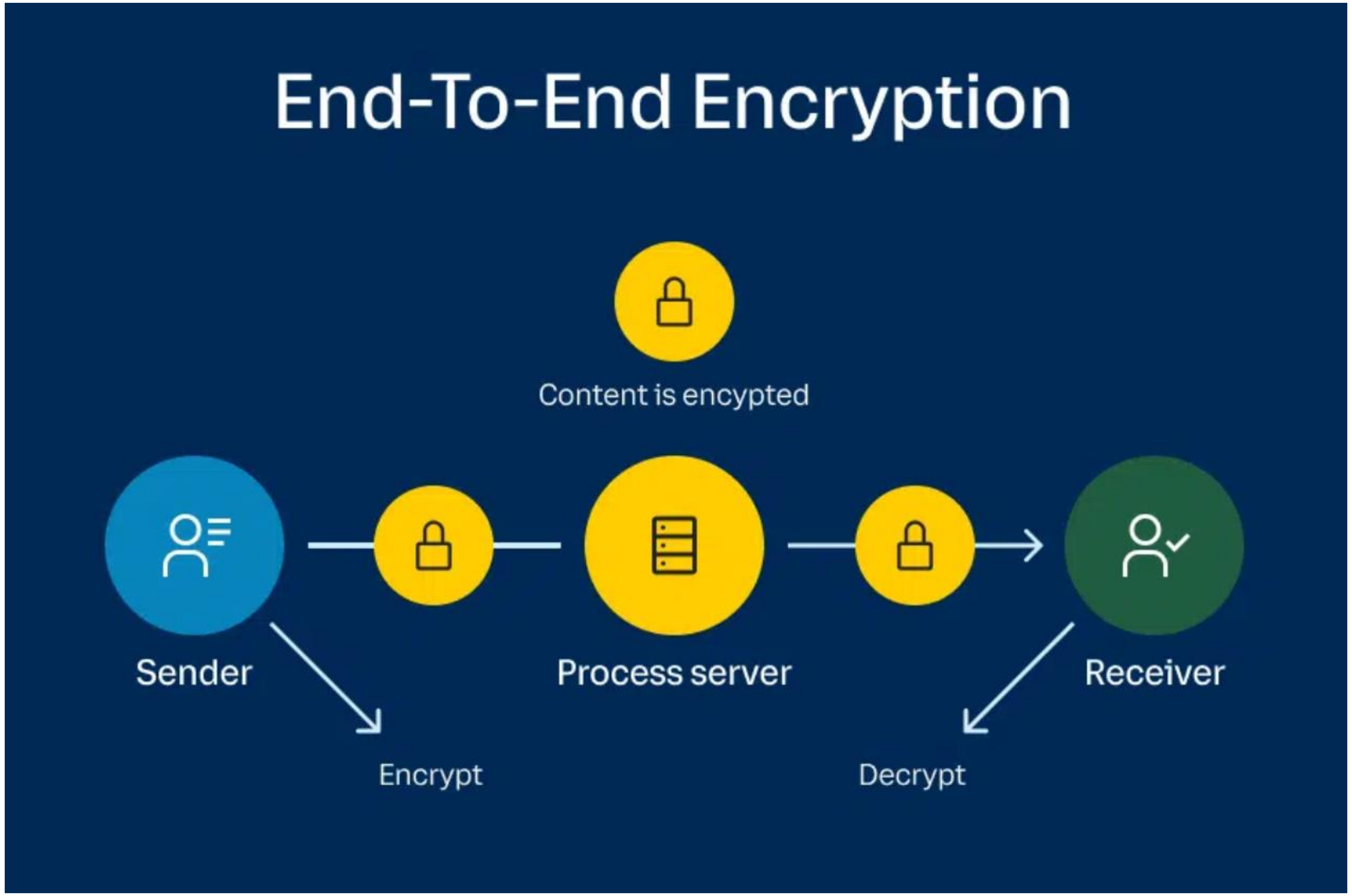
When the user asks: "Please help me check my account balance," the system collects only the account ID without requesting ID numbers or transaction history.

→

Without Data Minimization:
"Based on your loan record from July 2022, your loan limit is 500,000."
With Data Minimization:
"Your loan limit is 500,000."

Privacy Protection Techniques in NLP

End-to-End Encryption



1. **The sender** initiates communication and encrypts the message before sending it.
2. **The encrypted data passes through a processing server,** but the content remains **inaccessible** to unauthorized parties.
3. **The receiver decrypts the message using a private key,** ensuring that only the intended recipient can read the content.

Privacy Protection Techniques in NLP

End-to-End Encryption

Video Website: https://www.youtube.com/watch?v=c2OkOckSD20

Privacy Protection Techniques in NLP

End-to-End Encryption

A. Encrypted Model Inference for Sensitive Data

When users interact with LLMs using sensitive input data (e.g., legal queries, financial advice), E2EE can protect both the input and the output during model inference.

B. Secure LLM-Based Conversational Agents

In customer service chatbots and AI-powered virtual assistants, E2EE ensures that user queries and responses remain private during processing and storage.

Privacy Protection Techniques in NLP

End-to-End Encryption

Imagine a telehealth platform using an **LLM-powered virtual health assistant** that allows patients to consult doctors through an AI chatbot. Patients often share sensitive medical information such as symptoms, diagnoses, medications, and personal data like their name, ID number, or medical history.

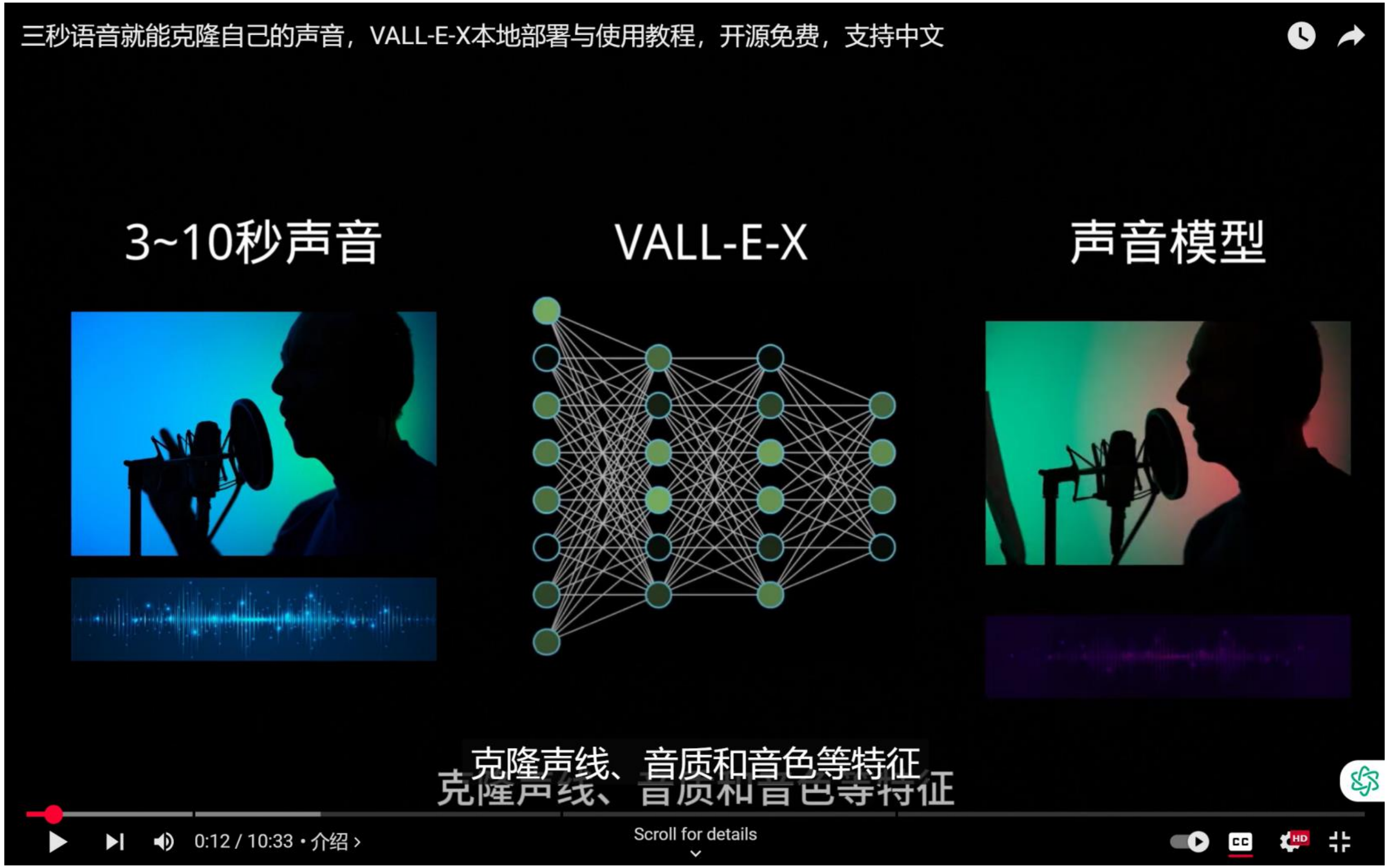
When a patient submits their query, for example: *"I have been experiencing severe chest pain and dizziness for the past two days. What could be the cause?"*, this message is immediately encrypted on the user's device before being transmitted.

→

The AI assistant's response—such as *"Your symptoms could indicate a cardiovascular issue. Please seek immediate medical attention."*—remains encrypted during transmission and is only decrypted once it reaches the user’s device.

VALL-E X

https://github.com/Plachtaa/VALL-E-X



Practice: Multilingual Text-to-Speech Synthesis and Voice Cloning

THE HONG KONG UNIVERSITY OF SCIENCE AND TECHNOLOGY (GUANGZHOU)

THE HONG KONG UNIVERSITY OF SCIENCE AND TECHNOLOGY

VALL-E X demo

Hello, my name is Nose. And uh, and I like hamburger. Hahaha... But I also have other interests such as playing tactic toast.

チュソクは私のお気に入りの祭りです。 私は数日間休んで、友人や家族との時間を過ごすことができます。

[EN]The Thirty Years' War was a devastating conflict that had a profound impact on Europe.[EN] [ZH]这是历史的开始。 如果您想听更多，请继续。[ZH]

AIAA 2290: Ethics, Privacy and Security in AI

Thanks!!

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The Hong Kong University of Science and Technology (Guangzhou)

2025 Fall